

Advancing MIMO Detection

Projected Gradient Descent for 16-QAM Modulation

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Outline

- Research Overview & Objectives
- Direction 1: Non-ML PGD - Methodology & Hyperparameter Analysis
- Direction 2: PGDNet - Architecture, Training & Learned Behavior
- Direction 3: DetNet - Implementation & Failure Analysis
- Comprehensive Results

The Detection Problem

y: Received signal (8 antennas)

H: Channel matrix (4×8 MIMO)

x: Transmitted symbols (16-QAM) (4 antennas)

w: AWGN

Channels: Kronecker, CDL-B (300ns delay, 10 - 30 km/h UE)

Noise Calibration: Adjusted to yield **LMMSE Baseline SER ~25%**

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w}$$

Objective: Near-optimal SER with practical complexity

Three Directions Tested

Non ML PGD

*Non Machine Learning
Projected Gradient Descent*

Algorithmic optimization with three projection strategies:

- None (baseline)
- 16-QAM (hard decision)
- Bbox (boundary box)

PGDNet

*PGD Network with Deep
Learning*

Deep unfolding approach with only 2 learnable parameters per layer:

- δ step size
- damping factor

DetNet ML

*Detection Network (Deep
MIMO Detection)*

Complex data-driven neural network with ~4,900 trainable parameters

Direction 1: Non-ML PGD Algorithm

$$\mathbf{x}_{\{k+1\}} = \text{Proj}(\mathbf{x}_k + \delta \mathbf{H}^H(\mathbf{y} - \mathbf{H}\mathbf{x}_k))$$

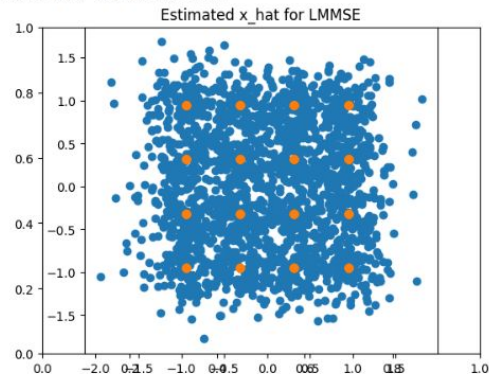
δ : Step size (learning rate)

Gradient ∇f : $\mathbf{H}^H(\mathbf{y} - \mathbf{H}\mathbf{x}_k)$

Projection Strategy Details

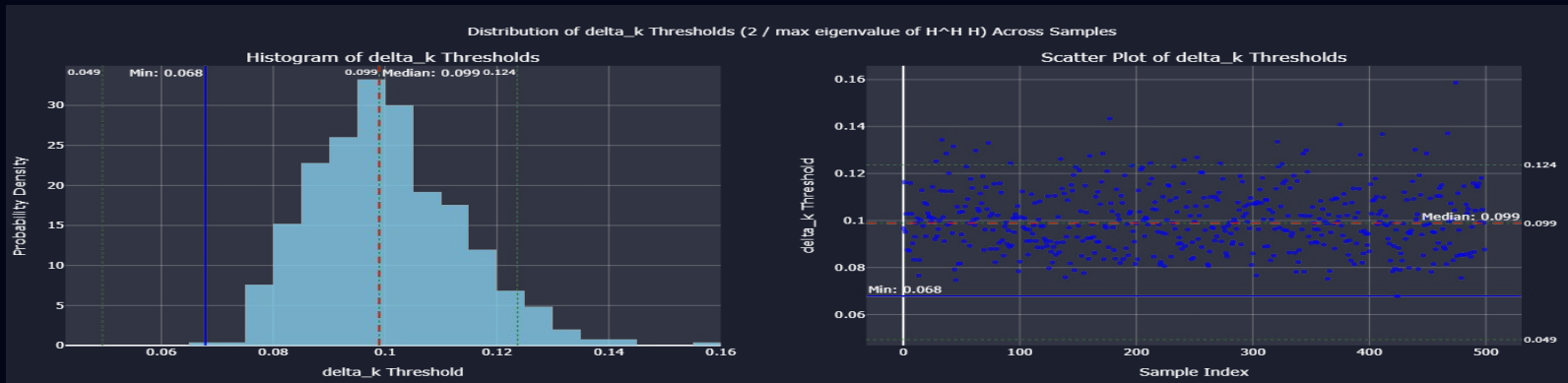
- **None:** $\mathbf{x}_{\{k+1\}} = \mathbf{x}_k + \delta \nabla f$ (no constraint)
- **16-QAM:** Hard decision to nearest constellation point, then damping interpolation
- **BBox:** Clip real/imag to $[-0.95, 0.95]$, damping factor 50-100%

Effective Noise Variance (LMMSE): 0.12877844
Theoretical Estimated Noise Variance (LMMSE): 0.12824634
Symbol Error Rate (LMMSE): 0.2575



Delta Threshold for Non-ML PGD

Critical Step Size Constraint



The optimal step size δ for Projected Gradient Descent is strictly bounded by the inverse of twice the largest eigenvalue ($\lambda_{\max}$) of $H^H H$.

Key Observation:

- Exceeding this threshold leads to divergence or oscillatory updates.
- Correct calibration of δ dramatically improves SER and convergence speed

For standardized comparison, manual delta values were tested alongside the optimal delta and damping parameters selected by Optuna.

Results on CDL Channel

Method	SER (Best)	Iterations	Optimal δ	Damping	Observations
LMMSE	25.75%	-	-	-	Baseline for comparison
16-QAM projection	25.5%	740+	0.015	0.05%	Low δ , low damping, delta = max from threshold
BBox projection	20.95%	160+	0.0677	50-100%	Aggressive + stable
None	28.7%	-	0.0677	-	Slow, high SER

Key Finding:

BBox achieves 5% improvement over LMMSE

BBox trajectory: Smooth descent, reaches <25% at ~100 iter

Damping paradox: Opposite optimal values (75 -100% vs 0.05%)

The observations from CDL channels generalize to Kronecker channel—BBox projection continues to outperform 16-QAM/None methods with similar hyperparameter sensitivity patterns

Direction 2: PGDNet Architecture and Training Configuration

Neural network with L layers, each layer = 1 PGD iteration

Only 2 trainable parameters per layer:

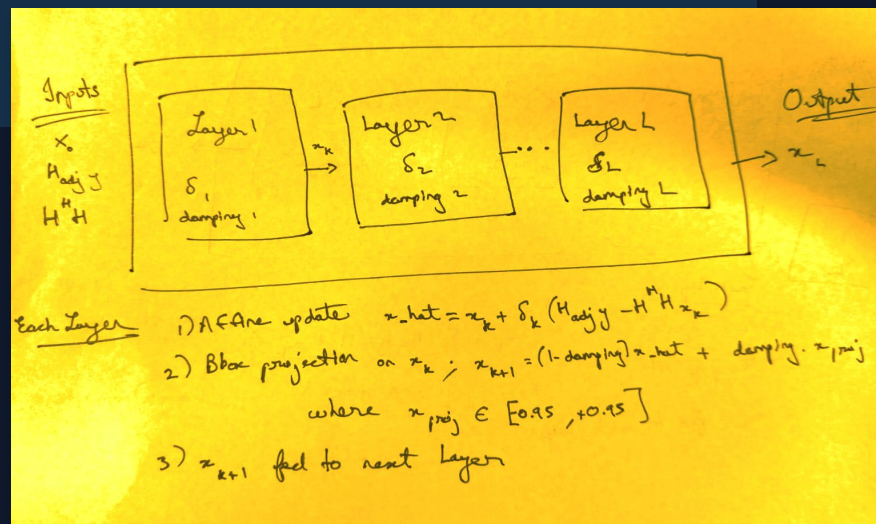
- δ (step size)
- damping_factor

Optimizer: Nadam (lr=0.012), 400+ steps, Batch 512

Loss: $\sum \log(k+2) \cdot \|x - x_h\|^2 / \|x - x_{ZF}\|^2$

Data: Real-time CDL generation, $\sigma^2 = 0.25$

Validation: Every 40 steps, early stopping enabled



Layer Depth Scaling Results

Layers	Val SER	Convergence	Parameters
5	~35%	Slow	10
10	~26%	Medium	20
15	~23%	Fast	30
20-30	~21%	~80 steps	40-60

Observation: Diminishing returns beyond 20 layers

Performance

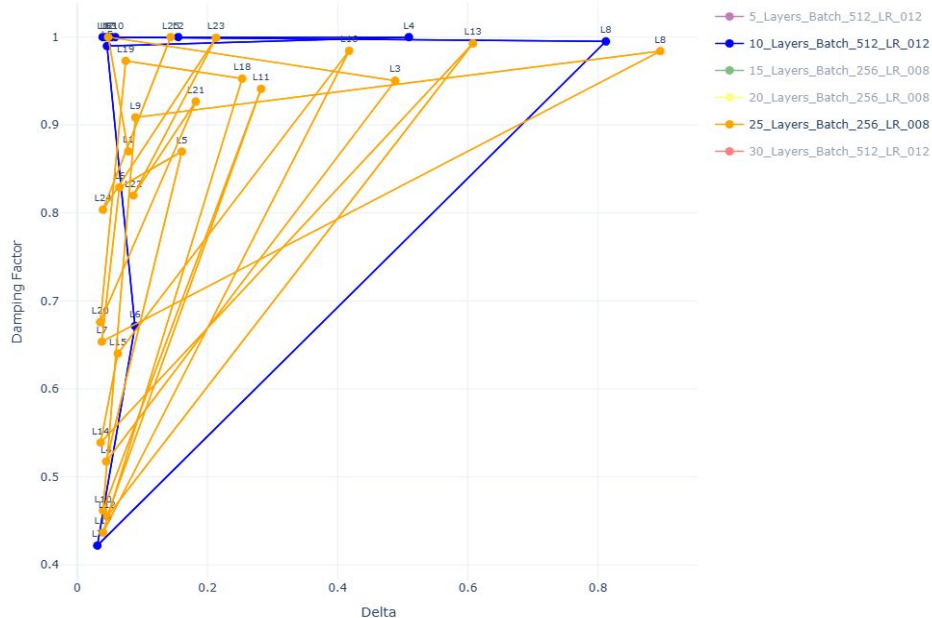


~20.5% SER achieved in 20-30 layers (vs 300+ iterations for Non-ML PGD)

Significant Speedup: 4-10× reduction in computational steps for same accuracy

Convergence Analysis

Delta vs. Damping Factor Across Layers



δ schedule: Non-monotonic! L8: 1.2, L15: 0.3, L22: 0.8

Damping schedule: Stays high (~ 1.0) even in final layers

Theory violation: Some $\delta > 1.0$ (beyond convergence bound)

Training stability: Loss plateaus at step ~ 80 , SER $\sim 21\%$

Learned Behavior Interpretation

Large δ strategy: Jump over flat regions, escape local minima

Variable δ pattern: Adaptive momentum-like behavior

High final damping: Forces hard decisions, acts as regularization

Open question: Theoretical guarantees for non-monotonic schedules

Direction 3: DetNet Implementation

Based on "Deep MIMO Detection" paper [Samuel et al.]

Architecture: 5 DetNetLayers, hidden_dim=16, fully connected

Forward pass: $x_k, v_k = \text{Layer}(x_{k-1}, v_{k-1}, H^H y, H^H H)$

Parameters: ~4,900 (W, b per layer) vs PGDNet ~60

Training: 800 steps, Nadam, batch 512, CDL channel

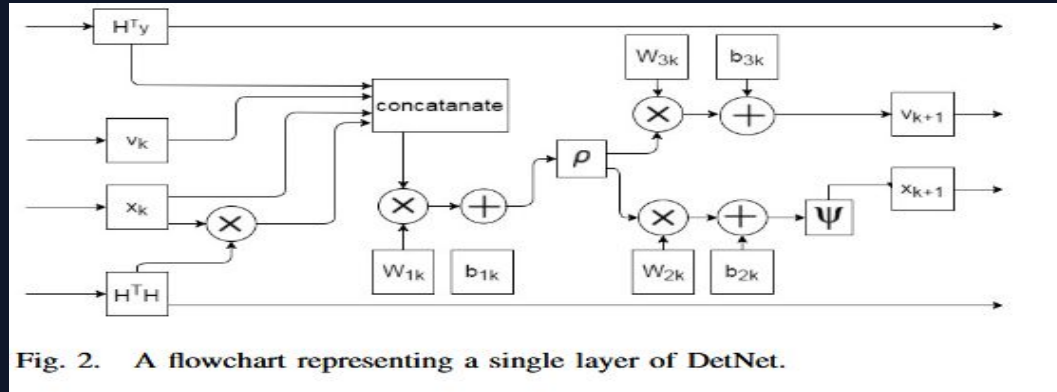


Fig. 2. A flowchart representing a single layer of DetNet.

Direction 3: DetNet Results

SER: ~34% (Worse than LMMSE!)

Very slow convergence, unstable training,

SER plateau: Stuck at ~34% after 800 steps

Conclusion: Not trivial to improve with more complexity

Comprehensive Benchmark (CDL, $\sigma^2=0.25$)

Detector	SER
Maximum Likelihood Detector	14.8%
K-Best (k=64)	15.75%
PGD-BBox	20.95%
PGDNet (25 L)	20.5%
LMMSE	25.75%
DetNet (5 L)	~34%

Methodology Conclusions

BBox projection: Aggressive δ (0.0677) + high damping (75-100%) optimal

16-QAM failure mode: Hard projection amplifies CDL multipath errors

PGDNet efficiency: 4-10 $\times$ iteration reduction vs non-ML PGD

DetNet lesson: 49 $\times$ more parameters $\neq$ better performance

CDL saturation floor: All PGD methods plateau ~20-21% SER

Model-driven superiority: Structure > parameter count

Future Research Directions

Immediate

- Multi-SNR training
- Higher-order QAM
- Larger MIMO systems

Thank You!

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